

JEE MAIN REVISION MASTER: WORK, ENERGY AND POWER 2026 2 QUESTIONS 3.33% WEIGHTAGE $\uparrow$ 99.4%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Work Energy And Power - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Work Energy And Power - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Work Energy And Power.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A horizontal force of 5 N is applied on a stationary body of mass 5 kg, which is initially at rest on a frictionless table. The change i..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Two bodies with kinetic energies in the ratio of 3:1 are moving with equal linear momentum. The ratio of their masses is..." [Ans: B]
- [2025 Shift] Q#3 (MCQ): "A body of mass 0.25 kg travels along a straight line from $x=0$ to $x=2m$ with a speed $v=kx^{3/2}$ where $k=2SI$ units. The work done by..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A horizontal force of 5 N is applied on a stationary body of mass 5 kg, which is initially at rest on a frictionless table. The change in kinetic energy of the body in 10 s is

- (A) 25 J
- (B) Zero
- (C) 125 J
- (D) 250 J

Correct Answer: None

Solution Analysis: The body is on a frictionless table, so the applied force produces acceleration without any loss due to friction. Given, $F=5\text{N}$ $m=5\text{kg}$ Using Newton's second law, $F=ma$ So, $a=(F / m)=(5 / 5)=1\text{m s}^{-2}$ The body starts from rest, so $u=0$ Time, $t=10\text{s}$ Using $v=u+at$ we get $v=0+1 \times 10=10\text{m s}^{-1}$ Initial kinetic energy is zero because the body is initially at rest. Final kinetic energy is $K=(1 / 2)mv^2$ $K=(1 / 2) \times 5 \times (10)^2$ $K=(1 / 2) \times 5 \times 100=250\text{J}$ So, the change in kinetic energy of the body in